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1  # MayaCardio - MAX30009 Design Objectives and Architecture Changes
2
3  # Part I - Design Objectives
4
5  ### O1 - Excitation Integrity
6
7  The delivered drive current equals the commanded current at every (current, frequency) in the ICG and BCA schedules on
8  a body-like load, and the board itself consumes only a small, known share of the compliance window.
9
10 ### O2 - Receive-Path Fidelity on Body-Like Loads
11
12 The sense path rejects the common-mode carrier that a person (not a resistor) produces, so impedance on a person is as
13 accurate as on a bench load.
14
15 ### O3 - Noise Floor: Broadband and Spurs
16
17 The board adds little to the chip's own noise in the ICG band and adds no discrete lines.
18
19 ### O4 - Calibration Accuracy and Traceability
20
21 Calibration is traceable, covers the clinical impedance band, and states which parts of the electrode path are inside
22 the correction.
23
24 ### O5 - Long-Duration Stability
25
26 Impedance on a fixed load is stable over a full clinical session after a short, declared warm-up, and drift can be
27 attributed.
28
29 ### O6 - Low-Noise Power Tree
30
31 Supply noise from the motherboard and isolated converter does not reach the AFE, digital activity does not modulate
32 AVDD/VREF, and the isolated rail has margin.
33
34 ### O7 - Layout and Grounding Integrity
35
36 The copper gives every analog signal a continuous return, keeps drive away from sense, keeps the sense pair symmetric,
37 and keeps aggressors away from the AFE.
38
39 ### O8 - Observability and Testability
40
41 The things we argue about can be measured on a built module without touching the CM4 or motherboard.
42
43 ### O9 - Patient Safety and Protection
44
45 Every electrode meets the single-fault DC limit, protection is designed and pulse-tested, and every new element that
46 crosses the isolation barrier is rated for it.
47
48 ### O10 - Compatibility and Drop-In Constraint
49
50 The V4 module is a drop-in replacement: same header and pin functions, no CM4 or ADS1293 change, and old firmware
51 cannot produce a silently wrong result on it.
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44
45 ---
46
47 # Part II – Architecture-Level Changes
48
49 ## A – Excitation and Drive Network
50
51 ### A1 – Drive Coupling Network and EXT_CAP Configuration
52
53 **Change (Architecture Level):**
54
55 Drive coupling network sized for EXT_CAP = 1 locked, across 1.5-500 kHz, with a stated settling budget.
56
57 **Design Comment:**
58 - 47 nF (DRVXC – SRVSJ).
59 - 220 nF (C38 and C39) will be changed.
60
61 **Objectives:** O1, O5
62
63 **Origin:** M-1; BCA RCA; roadmap RM7 superseded
64
65 **Class / MC:** MIN · MC-4
66
67 ---
68
69 ### A2 – Low-Frequency Drive-Loop Compliance
70
71 **Change (Architecture Level):**
72
73 Board share of the drive-loop burden kept small at the lowest BCA frequencies through drive-electrode coupling sizing.
74
75 **Design Comment:**
76
77 To be finalized by Suraj in the design.
78
79 **Objectives:** O1
80
81 **Origin:** BCA low-frequency DRV_OOR rows
82
83 **Class / MC:** MIN · MC-4
84
85 ---
86
87 ### A4 – Optional External Current-Setting Resistor
88
89 **Change (Architecture Level):**
90
91 External current-setting resistor footprint (DRVXR–DRVXC) retained as DNP.
92
93 **Design Comment:**
94

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95 Resistor footprint is available. We will have the option to test it later.
96
97 **Objectives:** O1
98
99 **Origin:** RM7
100
101 **Class / MC:** OPT
102
103 ---
104
105 ## B – Sense Input Protection and Symmetry
106
107 ### B1 – Low-Capacitance Sense Input Protection
108
109 **Change (Architecture Level):**
110
111 Specified, matched, low-capacitance, low-leakage sense protection. Single-fault series resistance kept at or above the
datasheet minimum and matched.
112
113 **Design Comment:**
114
115 To be finalized by Suraj in the design.
116
117 **Objectives:** O2, O9
118
119 **Origin:** M-2, M-11, RM5
120
121 **Class / MC:** MIN · MC-1
122
123 ---
124
125 ### B2 – Sense-Path Symmetry and Matched Routing
126
127 **Change (Architecture Level):**
128
129 Sense-path symmetry in parts and placement.
130
131 **Design Requirements:**
132 - Guard trace.
133 - Differential routing.
134 - Length-matched sense paths.
135
136 **Objectives:** O2
137
138 **Origin:** M-8, M-14
139
140 **Class / MC:** MIN · MC-1
141
142 ---
143
144 ## C – Driven Guard and Patient Cable Interface
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145
146 ### C1 – Driven Guard for Sense and Calibration Paths
147
148 **Change (Architecture Level):**
149
150 Driven guard from EL2A/EL3A specified as G-1...G-8 so that both EN_EXT_INLOAD and EN_INT_INLOAD are usable.
151
152 **Design Requirements:**
153 - Sense on EL2B/EL3B.
154 - EL2A/EL3A guard-only.
155 - Guard the whole BIP/BIN path, including CAL2/CAL3.
156 - Symmetric implementation.
157 - Stability footprint.
158 - Defined off-state.
159 - Shield links DNP.
160 - Small chip-side node.
161
162 **Objectives:** O2
163
164 **Origin:** M-4, datasheet p.48 and 0x42, Suraj Q6, RB-2
165
166 **Class / MC:** MIN · MC-1
167
168 ---
169
170 ### C2 – Shielded Patient Connector and Sense-Lead Guarding
171
172 **Change (Architecture Level):**
173
174 Patient connector with shield pins so sense-lead shields can be driven from the guard. Drive leads remain separate.
175
176 **Design Comment:**
177
178 To be finalized by Suraj in the design.
179
180 **Objectives:** O2, O3
181
182 **Origin:** A-2/RM10, 3 m cable
183
184 **Class / MC:** MIN (provision) · MC-1; cable REC
185
186 ---
187
188 ## D – Calibration Architecture and Switching
189
190 ### D1 – Precision Four-Wire Calibration Standards
191
192 **Change (Architecture Level):**
193
194 Precision four-wire standards on CAL1-CAL4.
195
```

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196 **Design Requirements:**
197 - 49.9  $\Omega$  lower standard.
198 - One upper standard sized by the RB-1 rule (~100  $\Omega$ ).
199 - Replace 680  $\Omega$  standard because DRV_OOR was measured at 640  $\mu$ A.
200
201 **Design Comment:**
202
203 To be finalized by Suraj in the design.
204
205 **Objectives:** O4
206
207 **Origin:** Suraj Q1/Q3; datasheet Fig. 16; RB-1
208
209 **Class / MC:** MIN · MC-5
210
211 ---
212
213 ### D2 – Electrode-Path Calibration Reference
214
215 **Change (Architecture Level):**
216
217 At least one calibration standard reachable through the electrode pins, so the board network stays inside a correction
or a fingerprint.
218
219 **Objectives:** O4, O5
220
221 **Origin:** Contract W8; Calibration package A7
222
223 **Class / MC:** MIN · MC-5
224
225 ---
226
227 ### D3 – Clinical Impedance Band Coverage
228
229 **Change (Architecture Level):**
230
231 Calibration standards must cover the clinically relevant impedance range.
232
233 **Design Requirements:**
234 - Lowest standard inside the chest  $Z_{\square}$  band.
235 - Upper standard at the top of the patient  $|Z|$  range in the patient configuration.
236 - Upper standard determined by measured DRV_OOR onset and the gain- $\times 10$  window.
237
238 **Objectives:** O4
239
240 **Origin:** Gap list item 2; RUN-D; RB-1
241
242 **Class / MC:** MIN · MC-5
243
244 ---
245

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246 ### D7 – Calibration Network Isolation and Architecture Comparison
247
248 **Change (Architecture Level):**
249
250 CAL network invisible to patient measurements.
251
252 **Design Requirements:**
253 - Same measurement configuration.
254 - CONNECT_CAL_ONLY = 1 with read-back.
255 - Relays default open.
256 - Guarded CAL2/CAL3.
257 - Connected-versus-removed A/B difference  $\leq 0.1\%$  /  $0.1^\circ$  (I-1...I-4).
258
259 **Design Comment:**
260
261 We will have the option to test two calibration architectures on the new design:
262
263 **Option 1 – CAL-Port-Based Calibration**
264 - Two resistors.
265 - One 2RC combination.
266 - Signal switches used for selecting the networks.
267
268 **Option 2 – Existing Relay-Based Calibration**
269 - Older multi-resistor calibration architecture.
270 - Relay-based selection of calibration resistors.
271
272 **Goal:**
273
274 Implement both options on the new design, test both, and select the architecture that makes the best contribution to
measurement accuracy and noise performance.
275
276 **Objectives:** O4, O2
277
278 **Origin:** Datasheet 0x41; RB-1
279
280 **Class / MC:** MIN · MC-5
281
282 ---
283
284 ### D4 – Reactive 2RC Verification Network
285
286 **Change (Architecture Level):**
287
288 Reactive verification network (2RC) switched onto the CAL port.
289
290 **Design Requirements:**
291 - 2RC network available through CAL switching.
292 - Characterised per unit.
293 - Intended for verification only.
294
295 **Design Comment:**

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296
297 To be finalized by Suraj in the design.
298
299 **Objectives:** O4
300
301 **Origin:** Suraj Q2/Q4; M-9
302
303 **Class / MC:** MIN · MC-6
304
305 ---
306
307 ### D5 – Fail-Safe Calibration and Patient Switching
308
309 **Change (Architecture Level):**
310
311 Calibration/patient switching only on the seven existing CM4 relay GPIOs (J3 → ULN2003).
312
313 **Design Requirements:**
314 - Use existing GPIOs.
315 - Fail-safe default state.
316 - Break-before-make switching.
317 - Barrier-rated switching elements.
318
319 **Objectives:** O4, O9, O10
320
321 **Origin:** Suraj Q3/Q5; RB-3
322
323 **Class / MC:** MIN · MC-7 · B9
324
325 ---
326
327 ## E – Power Supply, Grounding and Thermal Design
328
329 ### E1 – Power-Entry Filtering Before and After Isolation
330
331 **Change (Architecture Level):**
332
333 Module 5 V entry filtering to reduce conducted switching residue.
334
335 **Design Comment:**
336 - To be finalized by Suraj in the design.
337 - Filter placed before and after isolation.
338 - Input remains unchanged at 5 V.
339 - Converted to 3.3 V.
340
341 **Objectives:** O6, O3
342
343 **Origin:** M-5, RM3
344
345 **Class / MC:** MIN · MC-2
346
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347 ---
348
349 ### E2 – AVDD/DVDD Partitioning and Analog Ground Reference
350
351 **Change (Architecture Level):**
352
353 AVDD/DVDD partition, single-point AGND/DGND connection, and VREF returned to AGND.
354
355 **Design Comment:**
356
357 To be implemented as part of the PCB design.
358
359 **Objectives:** O6, O3
360
361 **Origin:** M-3, RM2
362
363 **Class / MC:** MIN · MC-2
364
365 ---
366
367 ### E6 – Thermal Separation and Heat Dissipation
368
369 **Change (Architecture Level):**
370
371 Heat sources separated from the AFE, with thermal copper used to manage temperature rise.
372
373 **Design Comment:**
374
375 On the MAX30009 daughterboard, apply good layout practices to increase heat dissipation so that thermal effects do not
    affect the sensor board.
376
377 **Objectives:** O5
378
379 **Origin:** M-12, RM13
380
381 **Class / MC:** MIN · MC-2
382
383 ---
384
385 ## F – PCB Stack-Up and Signal Routing
386
387 ### F1 – Multilayer PCB With Continuous Reference Planes
388
389 **Change (Architecture Level):**
390
391 Multilayer stack with an unbroken reference under the AFE, sense, and drive paths. No slot at the connector.
392
393 **Design Comment:**
394
395 To be implemented as part of the PCB design.
396
```

```
397 **Objectives:** 07, 03
398
399 **Origin:** M-8, RM1
400
401 **Class / MC:** MIN · MC-3
402
403 ---
404
405 ### F2 – Drive/Sense Separation and Noise-Aware Routing
406
407 **Change (Architecture Level):**
408
409 Drive and sense separated, sense pair matched, and aggressor lines kept away from the AFE.
410
411 **Design Requirements:**
412 - Physical separation of drive and sense.
413 - Matched sense pair.
414 - Relay control lines routed away from AFE.
415 - SPI lines routed away from sensitive analog paths.
416
417 **Design Comment:**
418
419 To be implemented as part of the PCB design.
420
421 **Objectives:** 07, 02
422
423 **Origin:** M-8
424
425 **Class / MC:** MIN · MC-3
426
427 ---
428
429 ## G – Observability, Diagnostics and Test Access
430
431 ### G1 – Delivered-Current Measurement and Test Access
432
433 **Change (Architecture Level):**
434
435 Delivered current measurement with zero added burden in use.
436
437 **Design Requirements:**
438 - End-of-line fixture measurement on every board using a ground-referenced precision load.
439 - In-use calibration-coefficient trending.
440 - 0 Ω Kelvin footprint in DRVP for DVT shunts.
441 - Original architecture description specifies no on-board amplifier.
442
443 **Design Comment:**
444 - To be implemented as part of the PCB design.
445 - Part finalized: AD8421.
446 - No isolation.
447
```

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448 **Objectives:** O8, O1
449
450 **Origin:** Suraj Q7, RM9, RB-4
451
452 **Class / MC:** MIN · MC-8
453
454 ---
455
456 ### G2 – INT/TRIG Isolation and Reserved I/O Connectivity
457
458 **Change (Architecture Level):**
459
460 Route INT (and optional TRIG) across the isolation barrier using an added isolator channel.
461
462 **Design Requirements:**
463 - Added isolator channel.
464 - Module J1 pins 3-7 (RSRV_IO).
465 - DNP links and test pads.
466
467 **Design Comment:**
468
469 To be implemented as part of the PCB design.
470
471 **Objectives:** O8
472
473 **Origin:** I-1, RM8, question 8
474
475 **Class / MC:** MIN · MC-8
476
477 ---
478
479 ### G4 – Analog Test Pads and Hardware Observability
480
481 **Change (Architecture Level):**
482
483 Provide test pads for critical analog signals and calibration nodes.
484
485 **Design Requirements:**
486 - Power rails.
487 - VREF.
488 - Drive nodes at the chip.
489 - Sense nodes at the chip.
490 - CAL nodes.
491 - Guard outputs.
492
493 **Design Comment:**
494
495 To be implemented as part of the PCB design.
496
497 **Objectives:** O8
498
```

```
499  **Origin:** O-5, RM12
500
501  **Class / MC:** MIN · MC-8
```